



Fig. 3: Model 4044 power sensor can be used in conjunction with 3141 Channel Power Monitor to measure the output power of either analogue- or digitally-modulated radios at power levels up to 100W



Fig. 4: Model 4045 directional power sensor can be used for forward and reflected composite power measurements at power levels up to 500W



Fig. 5: iVA-0627A cable and antenna analyser from Kaelus (Image courtesy: www.kaelus.com)

A new power measurement device launched recently serves the purpose of monitoring the components in analogue or digital LMR systems functioning between 144MHz and 960MHz.

Radio units, antenna, transmission line and power combiner are the usual elements that constitute an LMR. The channel power monitor from Bird Technologies monitors all elements of a radio system with the help of RF power sensors placed throughout the system. Apart from monitoring individual radios of a system, it also measures the system's total performance.

Engineers can monitor 16 channels, which can be expanded to accommodate new radio systems. They are alerted to measurement pa-

Make your mobile phone an RF meter

In sync with the IoT phenomenon, it is now possible to turn your Android phone into a power meter that would communicate with RF sensors to fetch power measurement data. Best suited for field engineers, Bird RF Meter app allows power measurements to be made on-the-go. With an intuitive and easy-to-use user interface, it enables engineers to set up sensors with appropriate settings and configuration options such as zeroing calibration, selection of element types, offsets and others.

The app automatically detects sensors and displays data of each sensor with serial numbers, making identification easier for engineers. In the current version, it shows readings for parameters like burst power, VSWR, crest factor, peak power, forward and reflected power (true average power) and complementary cumulative distribution function (CCDF). It is compatible with most USB field sensors from Bird Technologies.



A screenshot of the app while performing measurement (Image courtesy: www.birdrf.com)

rameters such as composite, forward and reflected power and out-of-spec conditions displayed and updated on a dedicated Web page generated by the model 3141 channel power monitor in real-time.

Data can be accessed on computers, tablets or mobile phones, which engineers can interpret to set alarms or configure the system to alert via email in case of failure conditions such as low/high power or poor-antenna voltage standing wave ratio (VSWR). Data logging capabilities further allow an engineer to view degraded performance and avoid emergency conditions.

Making RF measurements easier, quicker, more efficient

A notable current trend in network testing is the utilisation of simple, easy-to-use test equipment such that fundamental test behaviour and interpretation is not limited to professional RF engineers. There is also a growing use of analytics and cloud to boost efficiency in using such tools.

In line with this trend, this new analyser from Kaelus, equipped with

rechargeable lithium-ion batteries, reliably enables users to measure VSWR/return loss and the position of VSWR/return loss faults in their RF framework, thereby allowing estimation of distance to fault (DTF) in the RF path.

Bluetooth and universal serial bus (USB) capability in the iVA series cable and network analyser provides flexibility in measurement and opens new avenues of multi-port and sweep testing. It also allows engineers to make and view measurements by connecting this rugged device to the Bluetooth interface of a mobile phone, tablet, laptop or personal computer. There is an app particularly available for Android devices to view measurements on mobile devices.

The uncomplicatedness of this device significantly decreases test time on site as the site certification sweep testing process is made simpler. It is possible to directly connect this analyser to the device under test. This handy device can be used for interference checking in its spectrum monitor mode.